

Crude Drugs

• Definition

- Drugs are derived from the mineral, vegetable and animal kingdoms which consist of natural substances that have undergone no other processes than collection and drying.
- Crude Drugs are those natural products such as plants or part of plants, extracts and exudates which are not pure compounds.

- A botanical crude drug is from
 - Certain plant parts
 - Leaf or bark
 - The whole plant
 - Aerial part
 - Obtained from a plant, but no longer retains the structure of the plant or its organs
 - Volatile oils, Gum and resins
- Animals and insects
 - Honey, Cochineal (red dye)
- Microbials
 - Eg. Agar

Nomenclature of crude drugs

The following terms are used to indicate the parts of plants:

- Folia (or folium) = Leaf e.g. *Belladonnae folium* (Belladonna leaf) is the leaf of *Atropa belladonna* L.
- Semen = Seed e.g. *Strychni semen* (*Nuxvomica*) is the seed of *Strychnos nux-vomica* L.
- Flos = Flower. E.g. *Caryophylli flos* refers to flower buds of *clove* *Syzygium aromaticum*
- Fructus = Fruit. E.g. *Foeniculi fructus* refers to fruits of *funnel* *Foeniculum vulgare* Miller that are used for stomach and abdominal discomfort, as well as a spice in sweets and liqueurs.
- Cortex = Bark. E.g. *Cinchonae cortex* refers to the barks of several species of *Cinchona* that are sources of the antimalarial *quinoline alkaloids*.

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- **Radix** = Root e.g. *Liquiritiae radix* refers to roots of liquorice (licorice) *Glycyrrhiza glabra*
- **Rhizome** = Rhizome e.g. *Zingiberis rhizoma* refers to rhizomes of ginger *Zingiber officinale* Roscoe
- **Bulbus** = Bulb. E.g. *Allii sativi bulbus* refers to garlic bulb, *Allium sativum*
- **Herba** = herb. The crude drug consists of the aerial parts of the plant
 - e.g. *Echinaceae herba* refers to aerial parts of some members of the genus *Echinacea* that have immunostimulant behaviors.
- **Aetheroleum** = essential or volatile oil E.g. *Eucalypti aetheroleum* refers to *Eucalyptus* oil from *Eucalyptus globulus* Labill.
- **Lignum** = wood E.g. *Quassiae lignum* refers to woods of *Picrosma excelsa* and *Quassia amara* that have insecticide, anthelmintic, febrifuge and antimalarial activities.

Classification of crude drugs

1. Alphabetical

- Either Latin binomial or vernacular (common) names may be used.
- This arrangement is employed for dictionaries, pharmacopoeias, etc.
- While this method of classification may facilitate access to information, it offers the practitioner no therapeutic insights

2. Taxonomic

- On the basis of an accepted system of botanical classification, the drugs are arranged according to the plants from which they are obtained, in classes, orders, families, genera and species.
- It allows for a precise and ordered arrangement and accommodates any drug without ambiguity.
- As the basic botanical knowledge of pharmacy students decreases over the years this system is becoming less popular for teaching purposes.

3. Morphological

- Grouped according to shared anatomical features: leaves, flowers, fruits, seeds, aerial parts, herbs and entire organisms, rhizomes, bark, or roots.
- Such remedies are known in pharmacognosy as **organized drugs** because they are morphologically whole, or actual plant parts.
- **Unorganized drugs**, on the other hand, are plant materials that have no cellular structure, such as extracts, gums, resins, oils, fats, and waxes.
- These groupings have some advantages for the practical study of crude drugs; the identification of powdered drug is often based on micro-morphological characters.

4. Pharmacological or Therapeutic.

- This classification involves the grouping of drugs according to the pharmacological action of their most important constituent or their therapeutic use.
- However, it is important to appreciate that the constituents of any one drug may fall into different pharmacological groups (may have multiple pharmacological effects).

5. Chemical or Biogenetic.

- The important constituents (biochemical approach), e.g. alkaloids, glycosides, volatile oils, etc., or their biosynthetic pathways (biogenetic approach), form the basis of classification of the drugs.
- Ambiguities arise since one crude drug may contain a active compounds belonging to a different phytochemical groups.

Evaluation of crude drugs

- The analytical side of pharmacognosy is embraced in the expression “the evaluation of the drug,” for this includes the identification of a drug and determination of its quality and purity.
- By use of human senses to the application of advanced technology and equipment.
- These may be classified as follows: organoleptic, microscopic, biological, chemical, and physical

Organoleptic (Macroscopic) evaluation

- Organoleptic means "impression on the organs." and represents the simplest and the most human form of evaluation.
- Refers to evaluation by means of the *organs of sense*, and includes the *macroscopic appearance* of the drug (i.e. *sight*), its odor (*smell*) and *taste*, occasionally the sound or "*snap*" of its fracture (i.e. *hearing*), and the "*feel*" of the drug to the touch.
- For convenience of description the macroscopic characteristic of a drug may be divided into four headings, viz.: *shape and size*, *color and external markings*, *fracture and internal color*, and finally *odor and taste*.

Microscopic

Microscopic evaluation is valuable in the initial identification of herbs.

- It is mostly used for qualitative evaluation of organized crude drugs in entire and powder forms with help of microscope.
- It is also an indispensable technique for identifying the plant by characteristic tissue features, recognizing small fragments of crude or powdered herbs, or detecting adulterants or contaminants, such as insects, animal feces, mold, and fungi.
- Every plant possess a characteristic histology in respect to its organs (such as cellular tissues, trichomes, stomata, starch granules, calcium oxalate crystals and aleurone grains)
- Some characteristic features can be easily used to establish botanical identity and quality of the drugs.
 - Calcium oxalate crystals of Solanaceae species containing high amounts of atropine.
 - The characteristic glandular hairs of Lamiaceae and Asteraceae.



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- Different staining techniques are used to look for specific characteristics
 - Specific amino acids
- Quantitative aspects of microscopy includes study of stomatal number and index, palisade ratio, vein-islet number, size of starch grains, length of fibers etc which play important role in the identification of drug.

Physical

- In crude plant evaluation, physical methods, which rely on determination of physical constants, are often used to determine:
 - Solubility, specific gravity, melting point, water content, degree of fiber elasticity, optical rotation, refractive index, ash value and other physical characteristics.
- These are more usually applied to pure active principles rather than to the crude drugs.

Chemical

- Identification of the characteristic group or groups of active constituents that are already reported from the crude drug.
- Various chemical analysis methods are used to determine the identity and percentages of active principles:
 - Alkaloids, flavonoids, enzymes, vitamins, essential oils, fats, carbohydrates, protein or crude fiber-present.
 - Eg. **Borntrager's test** is commonly applied to all anthraquinone drugs.
 - Another example is a reaction that give acids and **flavonoids with Arnov's reagent** phenolic (the products of this reaction give purple color).
- All of these reactions are also used both for qualitative and quantitative analysis (colometric reactions).

- The final analytical process relies on more precise assays for determining quality.
 - Sophisticated techniques such as:
 - Gas chromatography (GC),
 - Thin layer chromatography (TLC), and
 - High-pressure liquid chromatography (HPLC) provide a chemical "fingerprint"
- These techniques are useful in identifying herbs, but also invaluable in standardizing extracts to chosen active ingredients or marker compounds.

Biological

- The biological evaluation– that is test for pharmacological activity, potency, and toxicity– of the plant drugs is one of the most important issues of pharmacognosy.
- There are plenty of methods for evaluation of biological properties of plant drug. For example,
 - Bacteria, such as *Staphylococcus aureus* are used to determine antiseptic value of the drugs (tolerable microbial range).
 - For standardization of *Digitalis* spp. (Foxglove) and other “heart tonic” drugs, pigeons and cats are used.

- Bioassay is the use of biological system to detect properties (i.e. pharmacological activity, potency, and toxicity) of a mixture or a pure compound.
- Bioassays could involve the use of:
 - *In vivo* systems (clinical trials, whole animal experiments),
 - *Ex vivo* systems (isolated tissues and organs),
 - *In vitro* systems (e.g., cultured cells).

